

The Pointing Role of Language

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Abstract

Language is often mentioned as a means of communication and knowledge transfer. However, its components and mechanisms are designed for a different function, namely, pointing at relevant phenomena, relying on the cognitive abilities of the communication parties. This paper provides arguments in favor of that claim and explains how language machinery performs that role.

Introduction

Humans are social animals. We need information transfer for learning purposes in childhood and coordination purposes in adulthood. Language is one of the tools used to facilitate those processes.

The role of language has been debated for centuries. Fodor [1] proposed a hypothesis that our thoughts use some form of language. Fedorenko et al. [2] considered two major roles, thought and communication, and based on their experimental findings, favored the latter one.

In this paper, I argue that the role of language in communication is limited, and it does not support information transfer. I propose to consider communication as a two-stage process. Language is involved in the first stage. However, information is not sent by the speaker. Information is received by listeners in the second stage using their perception, memory, or imagination.

To explain how language is used in the communication process, I analyze its components and mechanisms. This analysis consistently supports the main claim.

The Main Claim

I propose to consider communication as a two-stage process. The role of language is revealed during the first stage when a speaker uses language to guide a listener's attention to relevant phenomena in the current context. During the second stage, the listener's perception collects information about the relevant phenomena.

It is possible to refer to past or imaginary events. In those cases, listeners extract all the necessary information about relevant phenomena from memory or imagination.

Hence, the role of language is to point to or refer to relevant phenomena. Language does not encode and transfer information about them. It transfers a filter to be used by the listener's cognitive abilities to locate the relevant phenomena and collect required information about them.

Part I. Basics of intelligence

Interchangeability

According to Heraclitus [3], "No man ever steps in the same river twice, for it's not the same river and he's not the same man." Not only are all the objects unique, but the same object is different from what it was at any previous point in time or what it will be at any future point in time.

In practice, humans benefit from the interchangeability of some objects while respecting the uniqueness of others.

Referring to the "mind is a computer" metaphor [4-6], and taking into account that intelligence is about discerning [7], what kind of computation enables us to establish if objects are different? This brings us to understanding the nature of cognitive computation.

Nature of Cognitive Computation

I claim that cognitive computation relies on comparison. Not arithmetic, not matrix multiplication, not statistics, not first-order logic. Comparison is a kind of computation, but it's not the kind of computation you think about when someone mentions "computation."

We need to switch our analysis of cognitive functions to viewing them through the prism of comparison. This will speed up the progress of cognitive science.

Properties

Comparisons do not take as parameters the whole objects. They operate with properties. Objects have many properties. If we view properties as dimensions [8], then objects are multidimensional. Comparison can be viewed as a dimensionality-reduction operation.

Properties may be primitive, for example, height, weight, or color. They may be complex, for example, furniture, chairs, or fruits.

Concepts

What is important for interchangeability is comparability. Accurate measurements are not so important. Point-accurate measurements do not enable interchangeability, range-based estimates do. Ranges of comparable properties are the basis for concepts.

Rand [9] supports such a view, "Similarity is the relationship between two or more existents which possess the same characteristic (s), but in different measure or degree. The process of concept-formation consists of mentally isolating two or more existents by means of their

distinguishing characteristic, and retaining this characteristic while omitting their particular measurements, on the principle that these measurements must exist in some quantity, but may exist in any quantity. A concept is a mental integration of two or more units possessing the same distinguishing characteristic(s), with their particular measurements omitted."

Traditional views [10] do not rely directly on ranges of comparable properties. Showing the benefits of this approach is one of the goals of this paper.

Multidimensionality

Multidimensionality presents a challenge for processing [11]. Another challenge is the infinite number of options. The use of the finite number of ranges of comparable properties and dimensionality-reduction operations makes computations not only feasible but also fast.

Besides comparisons, there are other types of dimensionality-reduction operations - actions, categorization, references, etc. We will discuss them later in this paper.

Representation

Cognitive functions process objects in their multidimensionality. The results are reported using language, which relies on dimensionality reduction. This raises a need for proper representation of objects in memory. Existing approaches [12] do not enable the required flexibility.

There are several important details to understand about representations. Objects are multidimensional, and we should not store information about all the dimensions. First, we may be unaware of all of them. Second, we may consider some of them irrelevant for our purposes. Third, our memory is limited.

I propose a hypothesis that we store information about each of the relevant dimensions of an object separately. We store how a duck looks separately from how it quacks, and yet separately from how it swims. This raises a need to combine all those separate pieces together during retrieval. I propose to achieve that using some identifiers.

The above procedure may be extended to how we remember facts and scenes - we only store relevant pieces about them. This comes at a cost of losing details. This is reflected in [13], "When a memory is retrieved, all its fragments are put back together rarely identical to the initial experience."

The important aspect of such an approach to representation is that it enables us to manipulate objects in all their multidimensionality.

Core Algorithm

I hypothesize that the core algorithm of cognition also uses comparison. For any task, it picks available options that can serve as solutions, uses relevant constraints to filter those available options, and selects the most fitting option.

Options and constraints are chosen depending on the task. In the object recognition/categorization task, options are the known categories, and constraints are the object's properties. In a detective story, options are suspects, and constraints are the clues and alibis.

The proposed core algorithm of cognition relies on the ideas of the powerful binary search algorithm [14] and has logarithmic time complexity, which is suitable for real-time applications given that we use the abovementioned dimensionality-reduction operations and ranges that make the number of options to choose from finite.

Part II. Role of language in communication

Context

There are many definitions of context [15]. I support those that include all the objects available for our purposes. It is crucial that we do not include only relevant objects. It is crucial that we do not consider context as only the previous words in any given text. I insist on viewing context as belonging to the realm of objects. Even when we analyze texts, we either consider words and phrases as objects or we consider context as objects represented by those words.

The reason for the above requirement is that objects have multiple properties that make it possible for cognition to process them. Words, as dimensionality-reduction tools, do not have all of those properties.

Filtering

I propose a hypothesis that language does not refer to objects directly. Rather, it provides a filter for the current context. Objects caught by the filter are then what the reference is about. Traditional approaches to reference rely on truth conditions introduced by Frege [16]. In my approach, I depart from using truth conditions.

This is why we need to include all the objects in the context. References rely on the filtering procedure described above. If we include only relevant objects in context, we risk losing clarity of our references. Consider, for example, a shop with many kinds of balls. If our child likes only one of them, which makes it relevant, referring to it using a single word, "ball," will not help the shop owner to differentiate the required ball from the rest.

The filtering procedure does not take into account information that may be required for some purpose. It only needs the differentiating information. If someone asks to move a car, the current location of the car may or may not be used as a differentiating factor. However, for the task of "moving", it is definitely required.

Guiding Attention

Language does not transfer information. It passes a filter, which, applied against the current context, allows attracting a listener's attention to relevant objects. Then the listener's perception collects all the required information about those objects. Thus, communication is a two-stage process. Language participates in the first stage, when it guides a listener's attention. During the second stage, the cognitive abilities of the listener perform the cognitive heavy-lifting, collecting, validating, and processing the required information.

Michael Tomasello [17] mentions that utterances "express an intention that another share attention with them to some third entity."

Inviting Other Cognitive Functions

References to objects do not exhaust our uses of language. We also refer to actions, and what is more important, to connections of objects and actions - facts. In referring to the latter, we propose our view on those that can be confronted by listeners if they fail to validate that view against their experiences.

Language may engage other cognitive abilities besides attention and perception. For example, when talking about past events, it invites listeners to retrieve their memories of those events. When talking about imaginary events, we involve imagination and also reasoning to decide if those events are feasible.

Part III. Mechanisms of language. Generation

Referential Flexibility

Philosophers of language and linguists argue about the meaning of words and where it comes from, and how it undergoes the semantic drift. I introduce the concept of "referential flexibility" of symbols exactly to propose my view on how those processes occur.

That concept means that words, as special cases of symbols, easily acquire new meanings. Anyone can assign any meaning to any word. However, in order to efficiently use that word in that meaning in communication, that meaning for that word needs to be shared and agreed upon with the other party before a conversation. The more parties are involved, the harder it is to reach such an agreement.

Even if the agreement was reached, the subsequent use of that word in a new meaning has to pass the test of time. People have to make sure that it does not cause any ambiguity and that they are comfortable with it. One more aspect may relate to the need to refer to the underlying concept. If such a need diminishes, the word may be reassigned to mean something more worthy of talking about.

Meaning of Words

Words refer to the ranges of comparable properties. As such, they are abstract and do not refer to specific objects. We can view them as filters or dimensionality-reduction tools. They provide a lower-dimensional look at objects. I will discuss how this helps in referring to specific objects when we talk about references.

One of the issues with symbols is the grounding problem, introduced in Harnad [18]. The proposed hypothesis explains grounding by relating words to the ranges of comparable properties from the real world.

Ranges of comparable properties also provide an easy-to-understand approach to synonyms and antonyms. Synonym words refer to the same range of the same property. Consider the word "high," in its reference to the upper range of the "height" property, it is synonymous with "tall", in its reference to the upper range of the "voice pitch" property, it is synonymous with "feminine". Note that the words "feminine" and "tall" are not synonyms because they do not refer to the same range of the same property.

Many properties have the "average" range as a reference point. Antonyms refer to ranges of the same property symmetric with respect to that reference point. This is especially clear when we consider fine-tuned ranges like "warm" vs "cold", "hot" vs "freezing". Note that the same ideas are easily applied to verbs and modifiers.

When the same word refers to different ranges of different properties, it makes such a word polysemous. For example, the word "high" above is polysemous, as it refers to upper ranges of different properties - "height" and "voice pitch," to name a few.

Conventions

The idea that the meanings of words rely on conventions is not new. It was proposed and supported by many, for example, by de Saussure [19].

I want to attract attention to two aspects. The first one is about the size of the community. The smaller it is, the easier it is to introduce new terms and meanings. It is related to the cost of sharing the related information. The second one is about the scope of those conventions. Not only may the meanings of words be agreed upon, but also grammatical rules or the structure of longer texts may be regulated by such conventions.

Society

How a small community or society at large uses language depends on the needs of that group. As a result, the language may reflect how that community or society views the world. The respective hypothesis proposed by Whorf [20] was called "Sapir-Whorf Hypothesis" by Hoijer [21], even though the author called it "new principle of linguistic relativity":

"When linguists became able to examine critically and scientifically a large number of languages of widely different patterns, their base of reference was expanded; they experienced an interruption of phenomena hitherto held universal, and a whole new order of significances came into their ken. It was found that the background linguistic system (in other words, the grammar) of each language is not merely a reproducing instrument for voicing ideas but rather is itself the shaper of ideas, the program and guide for the individual's mental activity, for his analysis of impressions, for his synthesis of his mental stock in trade. Formulation of ideas is not an independent process, strictly rational in the old sense, but is part of a particular grammar, and differs, from slightly to greatly, between different grammars. We dissect nature along lines laid down by our native languages. The categories and types that we isolate from the world of phenomena we do not find there because they stare every observer in the face; on the contrary, the world is presented in a kaleidoscopic flux of impressions which has to be organized by our minds—and this means largely by the linguistic systems in our minds. We cut nature up, organize it into concepts, and ascribe significances as we do, largely because we are parties to an agreement to organize it in this way—an agreement that holds throughout our speech community and is codified in the patterns of our language. The agreement is, of course, an implicit and unstated one, but ITS TERMS ARE ABSOLUTELY OBLIGATORY; we cannot talk at all except by subscribing to the organization and classification of data which the agreement decrees."

Currently, the hypothesis has both supporters and critics [22], and more research is needed to show whether it is relevant.

References

Consider the context and relevant objects. The relevance may be related to any purpose. How do we use language to refer to the relevant objects in a given context?

We start with engaging our perception. We evaluate properties of the relevant objects and other objects in the context. We decide which properties differentiate the relevant objects from the rest. If one property is enough, we use that property. If one property is not enough, we use several properties. By properties here, I mean a variety of options - primitive properties, complex properties (categories), actions in which the relevant objects are involved, relations, etc.

What many experts call compositionality [23], I propose to view as stacked filters. This makes references context-dependent, because in different contexts the same set of filters catches different objects.

Note the use of the core algorithm of cognition. Options are objects in the context. Constraints are properties of the relevant objects. Filtering all the objects in the context leaves us with the relevant ones.

It is possible to refer to the same objects in the same context using different sets of filters. The set of filters required for differentiation may be different from the set of properties that make those objects relevant for some purpose. Also, any of those sets may be different from the defining features required for the categorization of those objects.

The most useful properties for differentiation are categories, because they already represent a set of filters.

Negation

Usually, for references, we use those properties of relevant objects that those objects have. But sometimes it is faster to narrow down the set of objects in the context by using those properties that relevant objects do not have. In such cases, we use negation. One famous example includes "without bad habits."

Compare this approach to the traditional one [24], where negation is considered as introducing an opposite concept. Contrary to that, I view negation as crossing out respective options.

References can be easily illustrated using Venn diagrams. In set operations, negation is represented by a set complement. This makes set operations the key mathematical functions for intelligence.

Declarative Sentences

Language can refer not only to objects but also to facts. There are two principal types of facts: about the static state or about the dynamic changes/interactions.

A useful metaphor for declarative sentences is "formulas." The above two types can be represented with formulas SVC (subject-verb-complement) and SVO (subject-verb-object) [25]. Constituents play the role of components in those formulas.

Questions

The most interesting potential of the above formulas is revealed when we ask questions. We can ask them about any constituent, making it unknown, and making the other constituents the keys for retrieval. Let us recall the modular organization of memory proposed above. If we can find the provided constituents stored along with the same Fact ID, then we may retrieve the answer from that fact.

At this moment, I propose to stop using the "formula" metaphor. The reason for that is that the answer is not calculated based on the known parameters, as we do in mathematical formulas. On the contrary, we use something similar to multiple-choice problems, compatible with the core algorithm of cognition. Options are the stored facts, and constraints are the provided constituents. There are several options for outcome here - many fitting facts, one fitting fact, or no fitting facts. The last one corresponds to the answer "I don't know." The second one allows for the only answer. And in the first case, we have a freedom of choice among fitting answers, when we may use additional factors to decide which of them is the most fitting.

Meaning of Sentences

Above, I stated that the meaning of words is abstract. Compared to that, the meaning of sentences against a given context is specific. It specifies relevant objects in that context and their interaction. If the question is interrogative, then we request information about the unknown constituent given the other known ones.

We can also say that meaning reflects the "aboutness" of a sentence [26]. The proposed approach explains how we achieve that.

It is important to note that sentences resolve to specific objects (with their identifiers) and facts (which also have identifiers). This is where we no longer need those specific words that were used in those sentences. Now we may even rephrase those sentences, leaving their meanings almost the same.

Part IV. Mechanisms of language. Understanding

Coherence Constraints

According to Rodd et al. [27], 80% of common English words have more than one meaning. How do we decide in which meaning each word is used? According to Firth [28], "you shall know a word by the company it keeps."

To explain how it is achieved, I refer to the core algorithm introduced above. It relies on the selection of the most fitting option from the available ones, respecting relevant constraints. Words in a sentence impose constraints on each other. This is what I mean by "coherence constraints." Let us consider how this is done on several practical examples.

Disambiguation

Ambiguity [29] is an essential feature of language, which is designed by our intelligence to help us address interchangeable phenomena. The reuse of words makes language efficient, but creates the problem of resolving their meaning in each particular sentence and context. In this section, I will show how the described language toolbox and the core algorithm of cognition solve that problem.

Recall that objects have multiple properties. Here is the key to solving polysemy - objects are multidimensional but not omnidimensional. The set of properties of different objects contains many properties, but not all possible properties.

Consider the phrase "high key". This reference refers to one object, which should have properties that correspond to both polysemous words "high" (height, sound, blood sugar level, etc.) and "key" (locking tool, sound, programming term, etc.). Among many possible objects, only "sound" fits both. Note that each word constrains the choice of meaning for the other.

Consider the following example of anaphora resolution and available candidates: "Stiven saw a car. It was red. It was yesterday. It was raining." We are interested in understanding each "it". All the available candidates are provided in the first sentence, some of them implicitly. Overall, there are four "objects" mentioned - Stiven, car, day (implicitly), and the event of seeing (indirectly). The property "color" is only available for the "car" object. The "time" property is only available for the "event" object. The "weather" property is available for the "day" object. This example also demonstrates coherence constraints in action.

Related to anaphora is cataphora. It is when a "relative reference" is used before anything is said about an "object". For example, "Because he was in a hurry, John took a taxi". We claim that "he" provides a lot of information to include an object into our set of available objects - "person", "male". Later, we add the "name" property to the entry. But the use of coherence constraints remains - "being in a hurry" is coherent with "taking a taxi".

Levesque et al. [30] proposed to prepare pairs of sentences with small differences that affect the resolution of an ambiguity contained in them. Consider the following two sentences from that dataset:

Joan made sure to thank Susan for all the help she received.

Joan made sure to thank Susan for all the help she had given.

How should "she" be resolved in each case? Is it Joan or Susan? In my theory, I propose to keep track of expected direct/indirect objects for each action. In this case, "thank smb for smth" expects something good, to which "help" generalizes well. If she received help, she should be the subject of "thank", otherwise, she should be the direct object.

Sometimes, information from the sentence is not sufficient for the disambiguation of grammatical structures and, therefore, constituents. Consider the example, "Visiting relatives can be annoying". Is it "visiting whom" or "visiting who"? Imagine now that the following sentence is mentioned in the context, "I don't like the climate there." The "visit" action may change, among others, the property "exposure to climate." Note the use of anaphoric "there" and that "don't like" resolves to the same range of the "attitude" property as "annoying". All these considerations support the version that I visited my relatives.

As an alternative, consider the following hint instead: "I don't like when my house gets crowded." In this case, we consider "visit" as changing the property "number of people in my house". "gets crowded" means the increase in that number. Therefore, this hint supports the version that relatives visited me.

Consider the following example: "Yesterday I watched a match with X. He hit the ball." Can you resolve the words "match" and "ball" correctly without knowing who X is? Now substitute Ronnie O'Sullivan for X. You will have a "snooker match" and a "snooker ball". Substitute X alternatively with Messi and you will get a "football/soccer match" and a respective ball.

The shorter the message, the more opportunities there are for ambiguities. But longer texts provide many constraints to perform disambiguation.

Interaction with Cognition

Resolution of ambiguities is only a starting point in many cases. Listeners may encounter a situation when the proposed connection between the relevant objects fails validation against the context. Note that the validation itself does not rely on language. It uses perception, reasoning, memory, etc.

When people communicate, the exchange of messages rarely happens without engaging other cognitive abilities. Memory and reasoning first come to mind but do not exhaust the list. Planning, theory of mind, imagination, etc., are often used when the parties compete with each other. Not only does an interrogated criminal have to conceal the true sequence of events, but also he/she has to come up with a believable story which will be hard to validate.

One interesting example of cognitive participation in a conversation is the so-called cooperative completion, when a listener completes a sentence for the speaker. It requires the listener to resolve the incomplete sentence and find the word that will coherently complete the sentence. Note that this process does not rely on prediction or statistical methods, even though from the outside the proposed word may seem random. Sometimes the proposed word is wrong in that it relies on an incorrect resolution of the sentence, but that only means that the speaker provided little hints for proper resolution.

Part V. Acquisition

From Simple to Complex

Babies are intelligent. They engage their core algorithm to figure out stuff from birth. They have to begin with ordering their sensory input, so it is natural that they first learn directly observable concepts. Presented with multiple red objects and hearing "red," they learn what color is and possibly many shades of the "red" range. Presented with cubes of different sizes and colors and hearing "cube," they learn what cubes are. They learn by figuring out or filtering a single common property, and ground a respective word that way.

It takes longer to figure out more involved concepts and map them to the respective words. For example, actions to verbs, abstract concepts.

Becker [31] mentions, "After a child has acquired (roughly) the initial fifty words, there is often (though not always) a sudden rush of word learning. This rush is like a burst of energy, or an explosion, in which the child seems to take a big step up in their word production. Typically, the first such vocabulary burst (and many subsequent ones too) is a nominal explosion, meaning that many nouns are learned in a short amount of time. But later in development, when the child has several hundred words in their productive vocabulary, there may be an explosion of verbs."

First, we see that nouns are acquired earlier than verbs. But how can we explain those "bursts"? Recall that a comparative property is an axis with ranges. It may take some time to figure out the first range and the axis, but all other ranges will go fast.

Michael Tomasello [17] mentions, "most children also have in their early language some so-called frozen phrases that are learned as holophrases but will at some point be broken down into their constituent elements, for example, Lemme-see, Gimme-that, I-wanna-do-it, My-turn, and many others.

...

She must be able either to "break down" or to "fill out" her holophrases so that she can express her communicative intentions in the more linguistically articulated way of adult speakers. Learning how to do this depends on the child's ability to comprehend not only the adult utterance as a whole, but also the functional role being played by the different linguistic elements in that whole. This is the beginnings of grammar."

Differentiation takes time. It requires multiple examples with different components, but the core algorithm does its job, and at some point, "breaking down" and "filling out" occur, and after that, the telegraphic stage begins.

Becker [31] mentions, "Significantly later in development (around age 2;0 to 2;6), once the child has about 400 words in their productive vocabulary, function words (like auxiliary verbs, agreement markers, certain modal verbs, the copula verb be, conjunctions, and articles) increase in number and frequency."

This means that a child starts figuring out how to point at facts. Observing facts and hearing familiar words with those function pieces, the child figures out the role of the word sequence, agreement markers, conjunctions, etc.

Becker continues, "Sometimes children's early words are bound to the immediate context. That is, the child might use a general noun like birdie to refer to a stuffed animal bird, not to birds in general. This is known as underextension: when a word refers to a class that is smaller than the target class. ...

Opposed to underextension is a phenomenon referred to as overextension. Overextension occurs when a child takes a word's meaning to include a larger class than in the adult grammar. ...

Overextension is usually based on some perceptual features ...

Unlike underextension, overextension errors are very common and are found crosslinguistically."

Using features, it is easy to fix both of those issues. Expanding the range will fix underextension, and introducing additional differentiating factors will fix overextension. The fact that "overextension errors are very common" is easy to explain - the formation of concepts is in the process, and many differentiating factors are yet to be added.

Becker mentions, "Together, these two problems add up to what is referred to as the mapping problem. Given the inherent variability between what we experience in the world and how we talk about those experiences with language, it's not reasonable to think the child will know exactly which aspect of an object or situation is being labeled with a particular word without some inductive constraints."

This is exactly the reason for using the core algorithm. On the first encounter, there are many options. On the second encounter, there are many, but different options. A simple set intersection eliminates and filters out non-fitting options. Later, when a new range for a category is mentioned, "That fruit is a peach," the category label filters out multiple options, making it easier to map the word to an object.

Becker also mentions the categorization problem, what features define a concept, and how to figure out those, mentioning the prototype theory along the way. I have already addressed that, so I will not repeat myself.

Becker raises another interesting question, "Before a child can even learn what words mean, they must know that words actually can be linked to individual meanings — that is, that words (can) have the property of being referential.¹ Where does this knowledge come from?"

Observing any situation, we try to store everything about it in our memory. Observing a ball and our mom pointing at it and saying "ball," we remember all those pieces along with a lot of irrelevant stuff. After multiple such scenes, we collect enough information to filter out everything irrelevant, leaving only the relevant connection.

The core algorithm is powerful, but the real world provides enough complexity for language acquisition to take a long time.

Conclusion

If by communication we mean the transfer of information, then language alone is a poor tool for that. I claim that the role of language is to point at relevant phenomena - objects, actions, and all the other constituents. One specific phenomenon that the language can point to is the connection of constituents into one fact.

The word "apple" does not convey all the information about the apple to which a speaker refers. That information is collected by the listener's perception after one's attention has been guided to the apple by the reference. Similar mechanisms engage the listener's memory or imagination.

The way a speaker connects constituents is subject to verification by the listener if it is possible against what one can perceive or check against one's experience. During the second stage of communication, the information lost in linguistic encoding is recovered using the listener's perception and verified.

Communication, thus, involves the speaker's utterances to guide the listener's attention to engage the latter's perception/memory/imagination to collect or recreate the relevant information.

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